

65. BETA-ALANINE FOR CYCLING by JEFF ROTHSCCHILD PhD

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What is Beta-Alanine?

Beta-alanine is a non-essential amino acid that serves as a building block for carnosine, a compound stored in muscles. **Unlike other amino acids, beta-alanine does not contribute directly to protein synthesis but plays a crucial role in buffering acidity in muscle cells. When ingested, beta-alanine combines with histidine to form carnosine, which helps maintain pH balance during high-intensity exercise.** This buffering effect is particularly useful in delaying muscular fatigue and sustaining performance.

Why is Beta-Alanine Used in Cycling and Endurance Sports?

In endurance sports like cycling, sustained high-intensity efforts (above second threshold/FTP/CP) generate hydrogen ions (H⁺), which lower muscle pH and contribute to fatigue. Carnosine, facilitated by beta-alanine supplementation, helps neutralize these hydrogen ions, potentially improving performance. **Beta-alanine is particularly beneficial in race scenarios involving efforts above the second threshold/CP/FTP (repeated accelerations/sprints, time trials, or climbs where the drop in pH can limit performance.**

The physiological mechanism and its area of application is very similar to what we saw ([here](#)) for sodium bicarbonate. The difference is that sodium bicarbonate acts as an extracellular buffer, while beta-alanine primarily enhances intracellular buffering capacity, making it complementary rather than redundant when used in addition to sodium bicarbonate.

Does Beta-Alanine Boost Race Performance?

Beta-alanine supplementation has been shown to improve performance in events lasting 1–10 minutes, where acidosis is a limiting factor (Saunders et al., 2017). **However, its effects on endurance events beyond 60 minutes are less clear, likely related to the varying nature of the exercise bouts.** For instance, one study found that beta-alanine supplementation significantly increased peak power output during sprints at the end of an endurance cycling race (Van Thienen et al., 2009), but another study indicated no improvement in short-duration sprints or time-trial performance during prolonged cycling (Perim et al., 2022). In the later study, the sprints were only 10 seconds in duration, and the time-trial was roughly 12.5 mins, both of which are outside of the optimal 1–10 min window where beta-alanine may shine.

Thus, its effectiveness may depend on the specific demands of the race. While it may not directly enhance prolonged endurance efforts, it could improve the ability to attack, surge, or sustain high power outputs during critical race moments.

For cyclists, beta-alanine may be particularly useful in:

- Short-to-medium-duration efforts (3–10 min)
- Repeated sprint efforts (e.g., criteriums, breakaways)
- Final sprint finishes after prolonged efforts

However, its effects are less pronounced in ultra-endurance efforts where neuromuscular fatigue and glycogen depletion rather than muscular acidosis is the primary limiter.

Does Beta-Alanine Improve Training Adaptations?

Beyond acute performance enhancement, beta-alanine may improve long-term training adaptations by allowing athletes to push harder during high-intensity sessions. Studies showing beneficial effects have typically not clamped training intensity, meaning athletes perform intervals based on effort, rather than pre-determined wattages (Rothschild & Bishop, 2020). As shown below, there may be additive effects between b-alanine and sprint interval training, likely due to higher workloads that can be completed with supplementation:

Average % improvement from baseline in 4km Time-Trial Performance in trained cyclists after 4 weeks of:

- Ingesting a placebo + regular training: +0%
- 4 weeks of beta-alanine supplementation + regular training: +1.5%
- 5 weeks of sprint-interval training + placebo: +2.8%
- 4-week loading phase of beta-alanine followed by 5 weeks of sprint-interval training with continued supplementation": +4%

*Reference: Bellinger & Minahan, 2016a; Bellinger & Minahan, 2016b; Rothschild & Bishop, 2020

However, overuse of beta-alanine in training should be approached cautiously. While it enables harder efforts, excessive high-intensity training without adequate recovery may lead to overtraining.

Dosage for Racing and Training

- **Racing:** Since beta-alanine requires a loading phase to be effective, acute supplementation on race day will have no immediate effect. Instead, maintaining elevated muscle carnosine levels over weeks/months is the key. So, using chronic supplementation during training periods until race day is the proper approach.
- **Training:** A daily dose of **3.2–6.4 g** of beta-alanine, divided into smaller doses (e.g., 800 mg x 4–8 times per day) over **4–6 weeks**, is recommended to maximize muscle carnosine stores for the race day.
- **Maintenance:** After the loading phase, a lower daily dose (~1.2 g) can help sustain elevated carnosine levels.

Side Effects

The primary side effect of beta-alanine is paresthesia, a tingling or itching sensation, particularly on the face, hands, and arms. While harmless, this can be uncomfortable for some athletes. Splitting doses into smaller amounts throughout the day and using a sustained release formula can help mitigate this effect.

What is the Best Form of Beta-Alanine?

Beta-alanine is available in various forms:

- **Standalone powder or capsules** – The most cost-effective and widely available form.
- **Pre-workout blends** – Often combined with caffeine and other ergogenic aids but may contain suboptimal beta-alanine doses.
- **Sustained-release formulas** – Designed to minimize tingling sensations while maintaining absorption.
- **Topical carnosine** – There is a topical product which shows promise (Sharpe & Macias, 2016), but research is still in its early days. Topical carnosine can potentially avoid the problems related to loading and side effects.

Does Combining Beta-Alanine with Other Supplements Enhance Its Effects?

Beta-alanine can be effectively combined with other ergogenic aids for synergistic effects, however research on co-ingestion is more challenging to do and so we have far fewer studies to look at. However, the potential for a ceiling effect also exists, meaning that beyond a certain point, additional supplementation may not yield further benefits. More research is needed to fully understand these interactions.

- **Sodium bicarbonate:** Works as an extracellular buffer, complementing beta-alanine's intracellular buffering effect. There may be a small additive benefit, most likely observed during supramaximal (i.e., above 100% peak power or $\text{VO}_{2\text{max}}$) exercise (Naderi et al., 2016).
- **Creatine:** May enhance power output and recovery when combined with beta-alanine, making it useful for sprint-heavy cycling disciplines (track cycling for example), but few studies are available.
- **Caffeine:** Since beta-alanine primarily affects buffering capacity rather than central fatigue, caffeine can provide an additional boost by enhancing alertness and reducing perceived effort.

- **Nitrates (beetroot juice):** While both beta-alanine and nitrates enhance performance through different mechanisms, their interaction is not well studied. Some theories suggest potential competition between buffering (through beta-alanine) and vasodilation (through nitrates) pathways, so it is not clear if they bring to additional positive effects when summed.

Summary

Beta-alanine is a well-supported supplement for endurance athletes, particularly those engaging in high-intensity, repeated efforts. While its effects on ultra-endurance performance remain inconclusive, it can be a valuable tool for cyclists seeking to improve time trial performance, repeated sprint ability, and race-day surges. To maximize its benefits, consistent daily supplementation is required over several weeks.

As with all supplements, individual responses vary, and it is best to trial beta-alanine in training before relying on it for competition.

Thanks for reading! Share with your Team!

Jeff Rothschild, PhD

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